

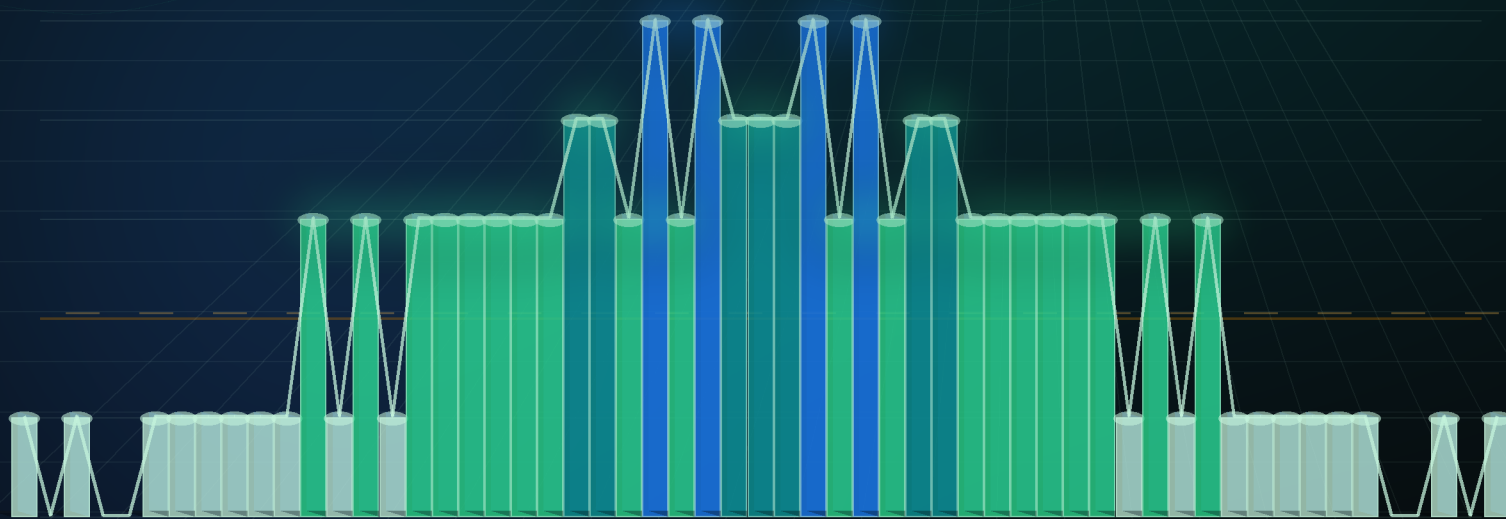
A SEVEN-ATOM CERTIFICATE

FOR THE CARDINAL-FUNCTION RANGE $\{1, 3, 4, 5\}$

A finite achievement set in which no point is
represented exactly twice

EXACT WITNESS

(2, 5, 6, 9, 10, 11, 13)



A Seven-Atom Certificate for the Cardinal-Function Range $\{1, 3, 4, 5\}$

A finite achievement set with no point represented exactly twice

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Research status

This manuscript gives a finite, exact certificate for entries printed as unknown in the summary table of Marchwicki and Žmija's 2025 preprint on bounded cardinal-function ranges. The finite algebra can be checked by direct multiplication or enumeration of 128 subsets. The result, its novelty, and the interpretation of the cited table have not yet undergone independent specialist review or peer review.

Abstract

For a finite sequence $x = (x_1, \dots, x_N)$ of positive integers, let $f_x(t)$ count the subsets of indices whose x -sum is t . A 2025 classification table of bounded cardinal-function ranges leaves open whether $\{1, 3, 4, 5\}$ can occur for a finite achievement set, for a Cantor achievement set, or without a prescribed topological type. We give the seven-atom witness

$$x = (2, 5, 6, 9, 10, 11, 13).$$

Its subset-sum generating polynomial has no coefficient equal to 2, while its positive coefficients assume each of the values 1, 3, 4, and 5. Hence

$$\text{rng}(f_x) = \{1, 3, 4, 5\}.$$

We state the exact coefficient identity, exhibit representative collisions, and verify the certificate independently by polynomial multiplication and exhaustive enumeration. Appending the uniquely represented tail $(3^{-n})_{n \geq 1}$ preserves every representation count and converts the finite achievement set into a Cantor achievement set. Subject to the cited table's conventions, the construction supplies affirmative entries in the finite, Cantor, and unrestricted classes. A bounded exact search record is included for reproducibility, but no claim of global minimality or publication priority is made.

Main conclusion. The sequence $x = (2, 5, 6, 9, 10, 11, 13)$ has cardinal-function range $\{1, 3, 4, 5\}$. Appending the uniquely represented tail $(3^{-n})_{n \geq 1}$ yields a Cantor achievement set with the same range.

MSC 2020. Primary 40A05; Secondary 05A15, 11B13, 11K31.

Keywords. Achievement set, cardinal function, subset sum, generating polynomial, Cantor set.

1 Problem context

Let $x = (x_n)$ be a finite sequence, or an absolutely summable sequence, of positive real numbers. Its *achievement set* is

$$\mathcal{A}(x) = \left\{ \sum_{n \in S} x_n : S \text{ is a subset of the index set} \right\}.$$

The associated *cardinal function* is

$$f_x(t) = \# \left\{ S : \sum_{n \in S} x_n = t \right\}, \quad t \in \mathcal{A}(x),$$

and its range is $\text{rng}(f_x) = \{f_x(t) : t \in \mathcal{A}(x)\}$.

Marchwicki and Žmija organize bounded ranges by the topology of the underlying achievement set [1]. Their notation includes:

- $\mathcal{F}$ for ranges realized by finite achievement sets;
- $\mathcal{C}$ for ranges realized by achievement sets homeomorphic to the Cantor set;
- $\mathcal{C}_v$ for ranges realized by Cantorvals;
- $\mathcal{R}$ for ranges realized without a prescribed topological type.

The authors state that blank entries in their summary table represent unknown cases. For

$$M = \{1, 3, 4, 5\},$$

the finite, Cantor, Cantorval, and unrestricted entries are blank, while two interval-type realizations are ruled out. The present construction addresses $\mathcal{F}$, $\mathcal{C}$, and $\mathcal{R}$; it does not settle the Cantorval entry.

Table 1: The classification entries addressed by this manuscript. “Blank” is the status printed in the August 2025 summary table.

Status for $M = \{1, 3, 4, 5\}$	$\mathcal{F}$	$\mathcal{C}$	$\mathcal{C}_v$	$\mathcal{R}$
2025 summary table	blank	blank	blank	blank
Candidate update proved here	yes	yes	unresolved	yes

The arXiv record inspected on 15 July 2026 lists the cited paper as version 1, submitted 18 August 2025. Targeted searches for the exact range and the witness below did not identify a later publication resolving these entries. Such a search cannot establish priority; it only motivates release of a compact, auditable candidate note.

2 Polynomial encoding of subset-sum multiplicity

The finite question is equivalent to a coefficient calculation.

Lemma 2.1 (Subset-sum coefficient identity). *Let $x = (x_1, \dots, x_N)$ be a finite sequence of positive integers. Then*

$$P_x(z) := \prod_{i=1}^N (1 + z^{x_i}) = \sum_{t=0}^{x_1 + \dots + x_N} f_x(t) z^t,$$

where $f_x(t) = 0$ for unattainable integers t .

Proof. In expanding the product, choosing 1 from the i th factor records omission of index i , while choosing z^{x_i} records its inclusion. Every subset contributes exactly one monomial $z^{\sum_{i \in S} x_i}$, so the coefficient of z^t is the number of subsets whose sum is t . $\square$

Remark 2.2 (Complement symmetry). If $T = \sum_i x_i$, subset complementation gives $f_x(t) = f_x(T - t)$. Hence P_x is palindromic. For the certificate below, $T = 56$.

3 The seven-atom certificate

Set

$$x = (2, 5, 6, 9, 10, 11, 13).$$

The coefficient calculation is made easier by splitting the atoms into two blocks:

$$u = (2, 5, 6), \quad v = (9, 10, 11, 13).$$

Each block has unique subset sums. Explicitly,

$$\begin{aligned} U = \mathcal{A}(u) &= \{0, 2, 5, 6, 7, 8, 11, 13\}, \\ V = \mathcal{A}(v) &= \{0, 9, 10, 11, 13, 19, 20, 21, 22, 23, 24, 30, 32, 33, 34, 43\}. \end{aligned}$$

Thus

$$\begin{aligned} P_u(z) &= \sum_{a \in U} z^a, \\ P_v(z) &= \sum_{b \in V} z^b, \end{aligned}$$

and every coefficient of $P_x = P_u P_v$ counts pairs $(a, b) \in U \times V$ with $a + b = t$.

Theorem 3.1 (Exact certificate). For $x = (2, 5, 6, 9, 10, 11, 13)$,

$$\text{rng}(f_x) = \{1, 3, 4, 5\}.$$

In particular, no point of $\mathcal{A}(x)$ has exactly two representations.

Proof. Direct multiplication of

$$P_x(z) = (1 + z^2)(1 + z^5)(1 + z^6)(1 + z^9)(1 + z^{10})(1 + z^{11})(1 + z^{13})$$

gives the exact grouped identity

$$P_x(z) = \sum_{t \in S_1} z^t + 3 \sum_{t \in S_3} z^t + 4 \sum_{t \in S_4} z^t + 5 \sum_{t \in S_5} z^t,$$

where

$$\begin{aligned} S_1 &= \{0, 2, 5, 6, 7, 8, 9, 10, 12, 14, 42, 44, 46, 47, 48, 49, 50, 51, 54, 56\}, \\ S_3 &= \{11, 13, 15, 16, 17, 18, 19, 20, 23, 25, 31, 33, 36, 37, 38, 39, 40, 41, 43, 45\}, \\ S_4 &= \{21, 22, 27, 28, 29, 34, 35\}, \\ S_5 &= \{24, 26, 30, 32\}. \end{aligned}$$

The six remaining exponents in $\{0, 1, \dots, 56\}$ are

$$1, 3, 4, 52, 53, 55,$$

and have coefficient zero. By [lemma 2.1](#), the positive coefficient set is therefore exactly $\{1, 3, 4, 5\}$. Since the range of the cardinal function is taken over attained sums, the zero coefficients are excluded. This proves the claim. $\square$

The identity has several immediate consistency checks. The four coefficient groups contain 20, 20, 7, and 4 exponents, respectively, so

$$20 \cdot 1 + 20 \cdot 3 + 7 \cdot 4 + 4 \cdot 5 = 128 = 2^7,$$

the total number of subsets. The sets are symmetric under $t \mapsto 56 - t$, as required by [remark 2.2](#).

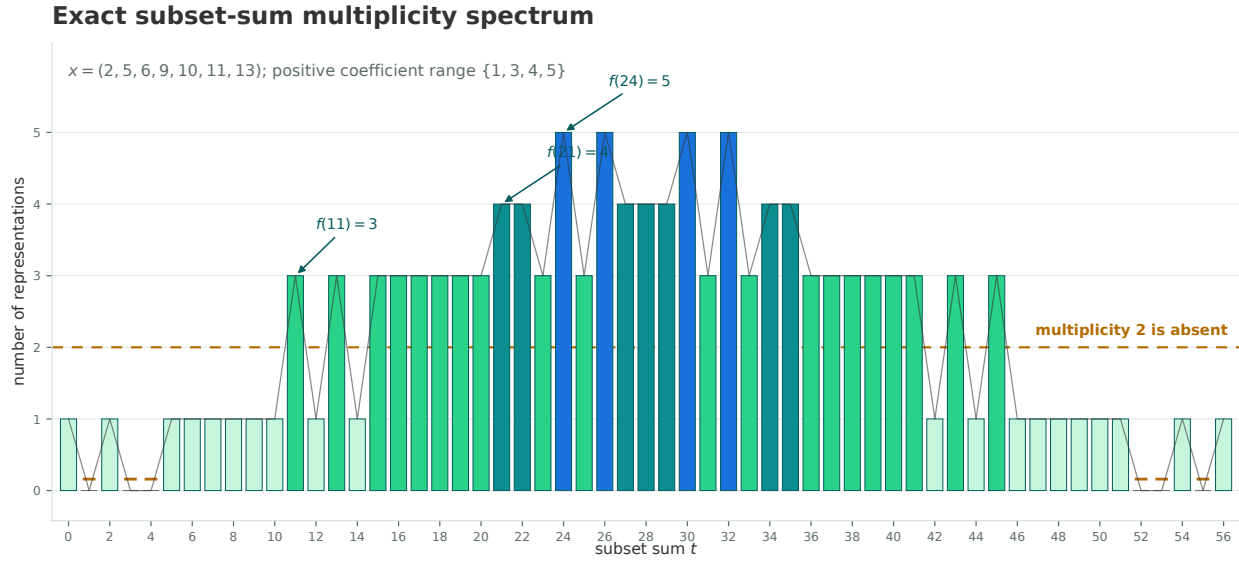


Figure 1: The exact coefficient spectrum. The dashed line at height two contains no coefficient; all four target multiplicities occur. Amber marks at height zero identify unattained integer sums.

3.1 Representative collisions

The multiplicities 3, 4, and 5 can be seen without expanding the full polynomial:

$$\begin{aligned}
 11 &= 11 = 5 + 6 = 2 + 9, \\
 21 &= 10 + 11 = 5 + 6 + 10 = 2 + 9 + 10 = 2 + 6 + 13, \\
 24 &= 11 + 13 = 5 + 9 + 10 = 5 + 6 + 13 \\
 &= 2 + 9 + 13 = 2 + 5 + 6 + 11.
 \end{aligned}$$

Thus $f_x(11) = 3$, $f_x(21) = 4$, and $f_x(24) = 5$. The grouped polynomial identity proves both that these lists are complete and that multiplicity two occurs nowhere.

4 A Cantor realization with the same range

The finite certificate can be converted into a Cantor achievement set without introducing any additional representations. Let

$$\tau = (3^{-1}, 3^{-2}, 3^{-3}, \dots)$$

and define

$$K = \mathcal{A}(\tau) = \left\{ \sum_{n=1}^{\infty} \frac{\varepsilon_n}{3^n} : \varepsilon_n \in \{0, 1\} \right\} \subset [0, 1/2].$$

The set K is a one-half scaling of the standard ternary Cantor set.

Lemma 4.1. *Every point of K has a unique representation by the sequence τ .*

Proof. Suppose two binary digit sequences first differ at index n . The absolute difference contributed at that index is 3^{-n} , while all later terms together have total

$$\sum_{k>n} 3^{-k} = \frac{1}{2 \cdot 3^n} < \frac{1}{3^n}.$$

Later digits cannot cancel the first difference, so distinct digit sequences give distinct sums. □

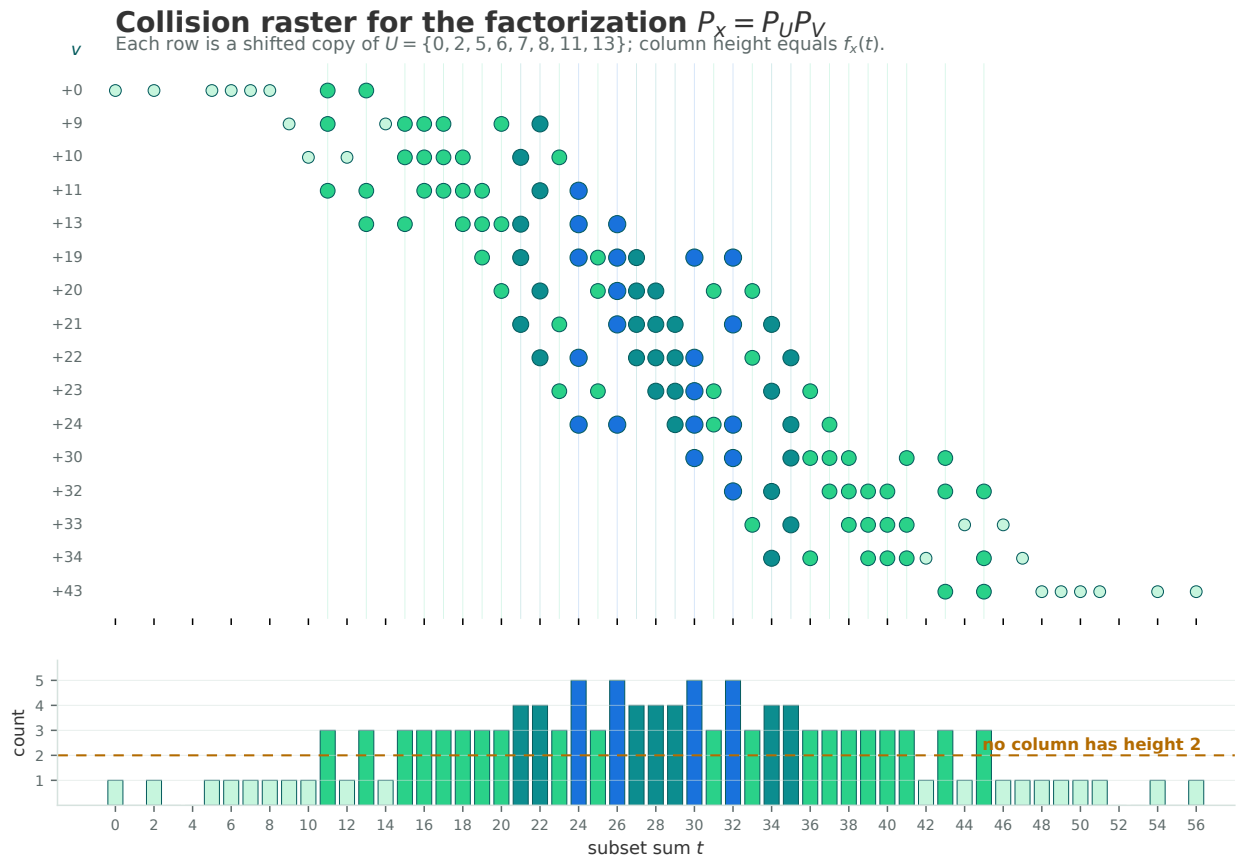


Figure 2: Convolution view of the certificate. Each row is one translate $U + b$, with $b \in V$. The number of points in a vertical column is the coefficient $f_X(t)$.

Let $x \sqcup \tau$ denote concatenation of the seven finite atoms with the infinite tail.

Proposition 4.2 (Cantorization). *The achievement set $\mathcal{A}(x \sqcup \tau)$ is homeomorphic to the Cantor set and*

$$\text{rng}(f_{x \sqcup \tau}) = \text{rng}(f_x) = \{1, 3, 4, 5\}.$$

Proof. Every point of the extended achievement set lies in

$$\mathcal{A}(x \sqcup \tau) = \bigcup_{s \in \mathcal{A}(x)} (s + K).$$

Distinct finite-prefix sums s and s' are distinct integers, hence $|s - s'| \geq 1$. Since $K \subset [0, 1/2]$, the translates $s + K$ and $s' + K$ are disjoint. Thus an extended sum determines its finite-prefix value s uniquely. By [lemma 4.1](#), it also determines its tail digits uniquely. Its only representation freedom is the selection of one of the $f_x(s)$ finite subsets summing to s . Consequently

$$f_{x \sqcup \tau}(s + k) = f_x(s) \quad (k \in K).$$

The cardinal ranges are equal.

Finally, $\mathcal{A}(x \sqcup \tau)$ is a finite disjoint union of translates of K . It is nonempty, compact, perfect, and totally disconnected, and therefore is homeomorphic to the Cantor set. $\square$

Corollary 4.3 (Classification update). *Under the notation of Marchwicki and Žmija,*

$$\{1, 3, 4, 5\} \in \mathcal{F} \cap \mathcal{C} \subset \mathcal{R}.$$

Thus the construction supplies affirmative finite, Cantor, and unrestricted realizations for the range $\{1, 3, 4, 5\}$.

Proof. The finite realization follows from [theorem 3.1](#); the Cantor realization follows from [proposition 4.2](#); and every finite or Cantor realization is, in particular, an unrestricted realization. $\square$

Cantor-tail extension preserves the finite multiplicity

The uniquely represented tail $K = \mathcal{A}((3^{-n})_{n \geq 1}) \subset [0, 1/2]$ is translated to each integer prefix sum.

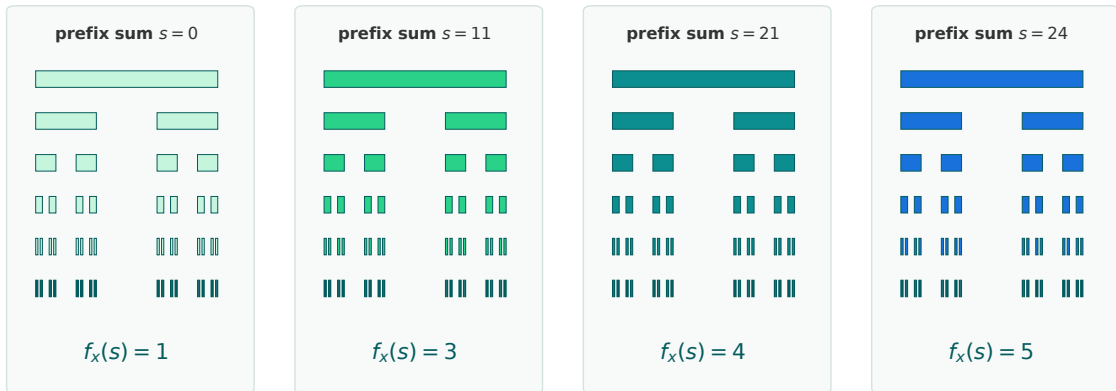


Figure 3: The uniquely represented Cantor tail is translated to each integer prefix sum. The component at s inherits exactly the finite multiplicity $f_x(s)$.

5 Discovery search and independent verification

The theorem is a finite symbolic certificate and does not depend on a computer once the seven atoms are stated. Computation was used to discover and independently audit the witness.

5.1 Bounded discovery search

The accompanying search script examines nondecreasing integer 7-tuples whose largest atom is at most 20. It multiplies the factors $(1 + z^i)$ exactly and applies two sound pruning rules:

1. coefficients never decrease as additional factors are multiplied, so a coefficient exceeding 5 cannot be repaired;
2. after an atom a has been appended, all future atoms are at least a , so coefficients of z^t for $t < a$ are frozen. A frozen positive coefficient outside $\{1, 3, 4, 5\}$ cannot be repaired.

After normalizing by greatest common divisor one, the search visits 75,812 recursive nodes and returns one solution in that bounded space:

$$(2, 5, 6, 9, 10, 11, 13).$$

This is a reproducible discovery record, not a proof that seven atoms are globally minimal or that the witness is unique without the stated bound.

5.2 Exact audit

A separate verification script computes the coefficient vector in two independent ways:

1. dynamic multiplication of all seven generating-polynomial factors;
2. direct enumeration of all 128 subsets.

It verifies agreement of the two calculations, the four coefficient groups in [theorem 3.1](#), the six unattained sums, complement symmetry, the coefficient total 128, the two unique block spectra U and V , the sample collisions at 11, 21, and 24, and the exact rational inequality used in [lemma 4.1](#).

Independent review target

The finite core can be audited by multiplying the two 0–1 polynomials P_u and P_v , or by checking the 128 subset sums. The only topological step is the disjoint-translate argument in [proposition 4.2](#). The principal non-algebraic review questions are prior art and whether the cited classification table is intended exactly as printed.

6 Scope and interpretation

The construction is intentionally narrow. It establishes a finite certificate and a Cantor extension for one bounded range. It does not determine whether the same range can be realized by a Cantorval, nor does it revisit the impossibility results for interval-type achievement sets stated in the source table.

The absence of the small certificate from a recent classification table is noteworthy but not logically problematic. Finite classification programs often leave isolated combinatorial cases that can be resolved by targeted exact search. The appropriate standard is not the apparent simplicity of the witness but whether the polynomial identity and class-transfer argument withstand independent review.

Priority and review caveat

The algebraic certificate is exact. The novelty claim is not independently certified. Equivalent witnesses may occur in unpublished notes, computational files, correspondence, or literature indexed under different terminology. Until specialists confirm both the argument and its priority, this document should be cited as a candidate classification update rather than as an established theorem of the field.

Reproducibility statement

The source package accompanying this preprint contains the complete Lua^AT_EX manuscript, formula-derived cover and figures, the exact verification script, the bounded discovery script, and recorded outputs. The finite calculations use integer arithmetic; the tail checks use exact rational arithmetic.

Declaration of generative AI and AI-assisted technologies

This manuscript is an exploratory research preprint issued by Metriq Foundation, Inc. The work was developed as part of Metriq’s experimental program in computationally assisted mathematics. OpenAI’s GPT-5.6 Pro was used as a research instrument to support problem selection, literature search, exact finite-certificate search, proof development, symbolic checking, mathematical exposition, figure design, and manuscript preparation.

Metriq Foundation directed the research request, determined the form and scope of the publication, and accepts responsibility for issuing the work as a candidate result. The manuscript has not yet undergone independent peer review. AI systems are not listed as authors, and independent verification, correction, and prior-art identification are expressly invited.

Data, code, funding, and competing interests

All source and verification materials are included with the release. No external dataset was used. This work received no external funding specifically designated for the result. The issuing organization declares no competing financial interest in the mathematical conclusion.

Suggested citation

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